

Binomial Distribution Guide

MDragon Data Tools - Study Guide

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1. What is a Binomial Distribution?

A **binomial distribution** describes the number of successes in a fixed number of independent trials, where each trial has only two possible outcomes (success or failure) and the probability of success remains constant.

Real-World Examples

- Number of heads in 10 coin flips
 - Number of defective items in a batch of 50
 - Number of students passing an exam out of 30
 - Number of free throws made out of 20 attempts
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2. Conditions for Binomial Distribution

For a binomial distribution to apply, **all four conditions** must be met:

1. **Fixed number of trials (n):** The experiment is repeated a specific number of times
2. **Two outcomes:** Each trial has exactly two possible outcomes (success/failure)
3. **Independent trials:** The outcome of one trial doesn't affect another
4. **Constant probability (p):** The probability of success is the same for each trial

Memory Tip: FTIC

- Fixed number of trials
 - Two outcomes
 - Independent
 - Constant probability
-

3. Formulas and Notation

Notation

- n = number of trials
- x = number of successes
- p = probability of success on each trial
- q = probability of failure = $1 - p$
- $P(X = x)$ = probability of exactly x successes

Binomial Probability Formula

$$P(X = x) = C(n, x) \times p^x \times q^{(n-x)}$$

$$\text{where } C(n, x) = n! / (x! \times (n-x)!)$$

Mean (Expected Value)

$$\mu = n \times p$$

Standard Deviation

$$\sigma = \sqrt{n \times p \times q}$$

Variance

$$\sigma^2 = n \times p \times q$$

4. Step-by-Step Examples

Example 1: Exact Probability

 **Problem:** A fair coin is flipped 5 times. What is the probability of getting exactly 3 heads?

 **Solution:**

1. Identify the values:

- $n = 5$ (number of flips)
- $x = 3$ (number of heads we want)
- $p = 0.5$ (probability of heads)
- $q = 0.5$ (probability of tails)

2. Check conditions:

- ✓ Fixed trials: 5 flips
- ✓ Two outcomes: heads or tails

- ✓ Independent: each flip independent
- ✓ Constant probability: $p = 0.5$ for all flips

3. Calculate combination:

$$\begin{aligned}C(5,3) &= 5! / (3! \times 2!) \\&= (5 \times 4) / (2 \times 1) \\&= 10\end{aligned}$$

4. Apply binomial formula:

$$\begin{aligned}P(X = 3) &= C(5,3) \times p^3 \times q^2 \\&= 10 \times (0.5)^3 \times (0.5)^2 \\&= 10 \times 0.125 \times 0.25 \\&= 10 \times 0.03125 \\&= 0.3125\end{aligned}$$

✓ Answer: 0.3125 or 31.25%

Example 2: Cumulative Probability (At Most)

 **Problem:** A basketball player has a 70% free throw success rate. If she takes 8 free throws, what is the probability she makes at most 5?

 **Solution:**

1. Identify the values:

- $n = 8$
- $p = 0.70$
- $q = 0.30$
- We need $P(X \leq 5)$

2. Calculate:

$$P(X \leq 5) = P(0) + P(1) + P(2) + P(3) + P(4) + P(5)$$

3. Using calculator or table:

$$P(X \leq 5) = 0.4482$$

✓ **Answer:** 0.4482 or 44.82%

Note: Use our Binomial Calculator for easier computation of cumulative probabilities

Example 3: Complement Rule Application

 **Problem:** A multiple choice test has 10 questions with 4 choices each. If a student guesses randomly on all questions, what is the probability of getting more than 3 correct?

 **Solution:**

1. Identify the values:

- $n = 10$ (questions)
- $p = 0.25$ (probability of guessing correctly)
- $q = 0.75$
- We need $P(X > 3)$

2. Use complement:

$$P(X > 3) = 1 - P(X \leq 3)$$

3. Calculate $P(X \leq 3)$:

$$P(X \leq 3) = P(0) + P(1) + P(2) + P(3)$$

4. Individual calculations:

$$P(0) = C(10,0) \times 0.25^0 \times 0.75^{10} = 0.0563$$

$$P(1) = C(10,1) \times 0.25^1 \times 0.75^9 = 0.1877$$

$$P(2) = C(10,2) \times 0.25^2 \times 0.75^8 = 0.2816$$

$$P(3) = C(10,3) \times 0.25^3 \times 0.75^7 = 0.2503$$

$$P(X \leq 3) = 0.7759$$

5. Apply complement:

$$P(X > 3) = 1 - 0.7759 = 0.2241$$

✓ Answer: 0.2241 or 22.41%

Example 4: Mean and Standard Deviation

 **Problem:** A factory produces light bulbs, and 5% are defective. In a sample of 200 bulbs, what is the expected number of defective bulbs and the standard deviation?

 **Solution:**

1. Identify the values:

$$\circ n = 200$$

$$\circ p = 0.05$$

$$\circ q = 0.95$$

2. Calculate mean:

$$\begin{aligned}\mu &= n \times p \\ &= 200 \times 0.05 \\ &= 10\end{aligned}$$

3. Calculate standard deviation:

$$\begin{aligned}\sigma &= \sqrt{n \times p \times q} \\ &= \sqrt{200 \times 0.05 \times 0.95} \\ &= \sqrt{9.5} \\ &= 3.08\end{aligned}$$

✓ Answer:

- Expected defective bulbs: 10
- Standard deviation: 3.08 bulbs

Interpretation: In a sample of 200 bulbs, we expect about 10 defective bulbs, with a typical variation of about 3 bulbs.

Example 5: At Least Probability

 **Problem:** The probability of rain on any given day is 0.3. What is the probability it rains on at least 2 of the next 7 days?

 **Solution:**

1. Identify the values:

- $n = 7$
- $p = 0.3$
- $q = 0.7$
- We need $P(X \geq 2)$

2. Use complement:

$$\begin{aligned}P(X \geq 2) &= 1 - P(X < 2) \\ &= 1 - [P(0) + P(1)]\end{aligned}$$

3. Calculate $P(0)$:

$$\begin{aligned}P(0) &= C(7,0) \times 0.3^0 \times 0.7^7 \\&= 1 \times 1 \times 0.0824 \\&= 0.0824\end{aligned}$$

4. Calculate $P(1)$:

$$\begin{aligned}P(1) &= C(7,1) \times 0.3^1 \times 0.7^6 \\&= 7 \times 0.3 \times 0.1176 \\&= 0.2471\end{aligned}$$

5. Apply complement:

$$\begin{aligned}P(X \geq 2) &= 1 - (0.0824 + 0.2471) \\&= 1 - 0.3295 \\&= 0.6705\end{aligned}$$

✓ Answer: 0.6705 or 67.05%

5. Using the MDragon Binomial Calculator

Steps:

1. Navigate to **Calculators** → **Binomial Distribution**
2. Enter the following parameters:
 - **n**: Number of trials
 - **p**: Probability of success (as a decimal)
 - **x**: Number of successes
3. Select calculation type:
 - **Exactly (=)**: $P(X = x)$
 - **At most ($\leq$)**: $P(X \leq x)$

- **At least ($\geq$):** $P(X \geq x)$
- **Less than ($<$):** $P(X < x)$
- **Greater than ($>$):** $P(X > x)$

4. Click **Calculate** to see:

- Probability result
 - Mean and standard deviation
 - Visual distribution graph
-

6. Practice Problems

Problem 1

A survey shows 60% of people prefer chocolate ice cream. If 12 people are randomly selected, what is the probability that exactly 8 prefer chocolate?

Problem 2

A quality control inspector finds that 3% of items are defective. In a batch of 50 items, what is the probability of finding:

- a) Exactly 2 defective items?
- b) At most 1 defective item?

Problem 3

A student has a 75% chance of correctly answering each question on a true/false quiz with 15 questions. What is the expected number of correct answers and the standard deviation?

Problem 4

The probability a seed germinates is 0.85. If 20 seeds are planted, what is the probability at least 18 germinate?

Problem 5

A basketball player makes 80% of her free throws. In the next 10 free throw attempts, what is the probability she makes between 7 and 9 (inclusive)?

Answers to Practice Problems

Problem 1

$$\begin{aligned}n &= 12, x = 8, p = 0.6 \\P(X = 8) &= C(12, 8) \times 0.6^8 \times 0.4^4 \\&= 495 \times 0.0168 \times 0.0256 \\&= 0.2128 \text{ or } 21.28\%\end{aligned}$$

Problem 2

a) Exactly 2 defective:

$$\begin{aligned}n &= 50, x = 2, p = 0.03 \\P(X = 2) &= 0.2578 \text{ or } 25.78\%\end{aligned}$$

b) At most 1 defective:

$$P(X \leq 1) = P(0) + P(1) = 0.5553 \text{ or } 55.53\%$$

Problem 3

$$\begin{aligned}n &= 15, p = 0.75 \\ \mu &= 15 \times 0.75 = 11.25 \text{ questions} \\ \sigma &= \sqrt{15 \times 0.75 \times 0.25} = 1.68 \text{ questions}\end{aligned}$$

Problem 4

$n = 20, p = 0.85$
 $P(X \geq 18) = P(18) + P(19) + P(20) = 0.4049 \text{ or } 40.49\%$

Problem 5

$n = 10, p = 0.8$
 $P(7 \leq X \leq 9) = P(7) + P(8) + P(9) = 0.6778 \text{ or } 67.78\%$

Quick Reference

Type	Notation	Formula
Exactly	$P(X = x)$	$C(n,x) \times p^x \times q^{(n-x)}$
At most	$P(X \leq x)$	Sum from 0 to x
At least	$P(X \geq x)$	$1 - P(X < x)$
Mean	μ	$n \times p$
Std Dev	σ	$\sqrt{n \times p \times q}$

Use our Binomial Calculator at MDragon Data Tools for quick and accurate calculations!

